

How to be Choosy: Guide to Preparing Big Datasets for Classroom Use

	Description		Pedagogical Benefits and Drawbacks	GAISE II and IDSSP Connections
Reduce Cases	Random	Cases are randomly removed.	+ Retain dataset's aggregate characteristics. - May lose pedagogically useful or interesting cases (e.g. outliers, known cases).	<i>GAISE II</i> Formulate Statistical Investigative Questions Analyze the Data <i>IDSSP</i> Basic Tools for Exploration and Analysis Graphs and Tables
	Purposeful	Cases are filtered by an available attribute, so data includes only cases with relevant characteristics.	+ Focuses on cases students know more about, or that will address specific questions. - Introduces sampling bias.	<i>GAISE II</i> Analyze the Data Interpret Results <i>IDSSP</i> Basic Tools for Exploration and Analysis Avoiding Being Misled by Data
	Build-Your-Own	Cases are manually or programmatically filtered based on implicit or composite attributes.	+ Focuses on issues defined by the teacher or students. + Motivates "data moves." - Difficult to assess for bias in the resulting dataset.	<i>GAISE II</i> Formulate Statistical Investigative Questions Consider Data <i>IDSSP</i> The Data-Handling Pipeline
Reduce Attributes	Thematic	Attributes are organized into conceptually related groups; only some groups are available to students.	+ Emphasizes modeling and theory-driven analysis. + Highlights data context. - May be used for illustration of target concepts rather than for inquiry. - Emphasis on theory may encourage focus on spurious relationships.	<i>GAISE II</i> Consider Data Analyze the Data Interpret Results <i>IDSSP</i> Basic Tools for Exploration and Analysis
	Mathematical	Attributes are selected and grouped based on quant. characteristics (such as data type) or associations (e.g. exhibiting a certain type of relationship).	+ Emphasizes target analyses, visualizations, models, relationships. + Highlights precision and detail in patterns and relationships. - May reduce students' focus on data context and inquiry process.	<i>GAISE II</i> Analyze the Data Interpret Results <i>IDSSP</i> Basic Tools for Exploration and Analysis
	Question-Driven	Attributes are grouped into collections that permit exploration of different student-generated questions.	+ Emphasizes data investigation cycle. + Context for prediction & hyp testing. - Students may require scaffolding to pose appropriate questions. - Difficult to anticipate the skills and models needed to pursue student-posed questions.	<i>GAISE II</i> Formulate Statistical Investigative Questions Consider Data <i>IDSSP</i> The Data-Handling Pipeline